

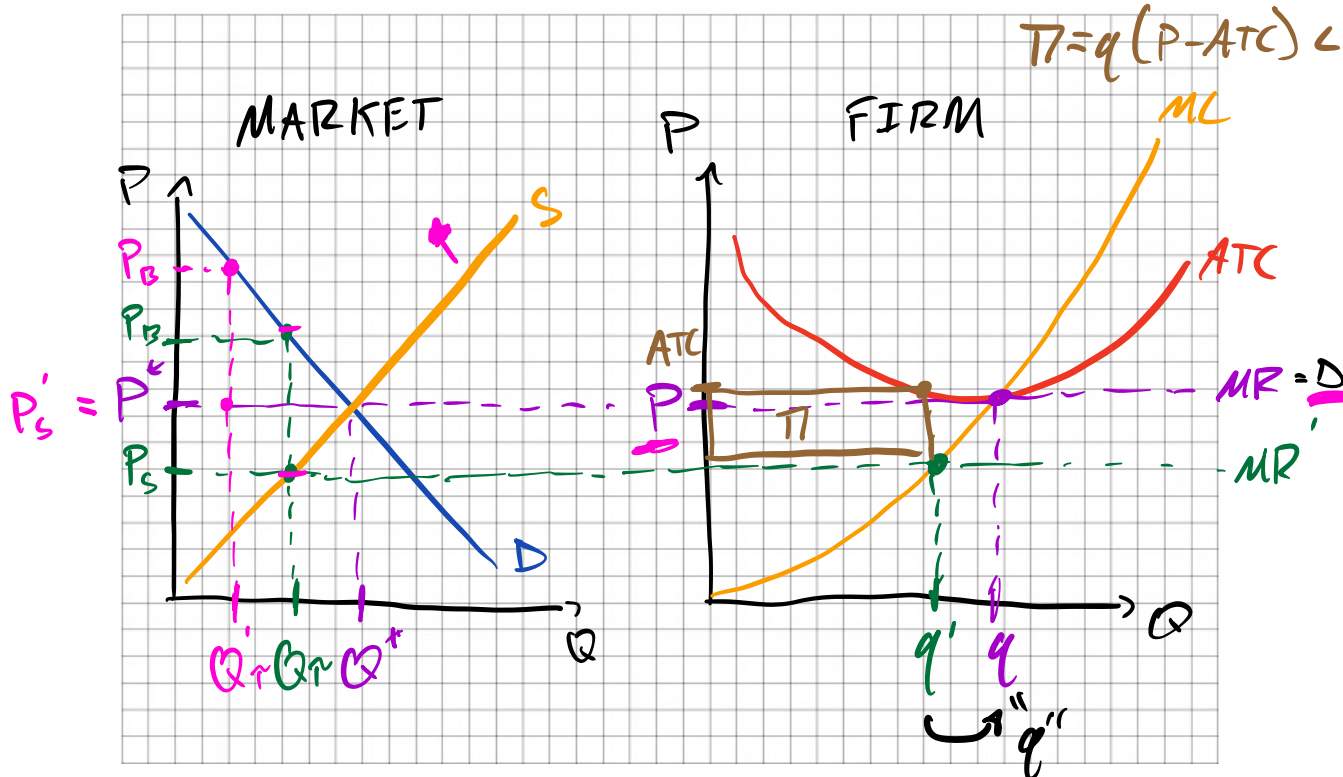
Honors Intro Micro | Homework 4 Demo

Question 1 | Taxing Cauldrons

Because of regulations imposed by the Ministry of Magic, all cauldrons had to be manufactured the same in every dimension. The hope was that this uniformity in production would make it easier to train witches and wizards to use the cauldrons, but an unexpected benefit was that the extensive specification documents served as an instruction manual for anyone wishing to begin a new cauldron manufacturing facility. Anyone could easily start manufacturing cauldrons, though the number of cauldron manufacturers had been constant over the past couple of years. This changed when the Ministry imposed a tax on all cauldron sales. Using a couple of graphs to illustrate your answer, explain what happened in the cauldron market because of this tax.

Loss ↘

$$\pi = q(P - ATC) < 0$$



SUMMARY

1. $\pi = 0$

2. $Q \uparrow < Q^e$, $q' < q$

3. $Q' \uparrow < Q \uparrow$, $q'' > q'$

- We start in long run equilibrium in a perfectly competitive market.

1. The tax lowers the seller's price leading to a loss.
- This loss pushes firms out of

the market.
 • In the long run this leads to a lower Q^* and zero profits.

Question 2 | Monopoly!

Wizard's Chess is a somewhat common game played by witches and wizards, similar to chess but with self-moving pieces. Due to its proprietary production process the game can only be obtained at Weasley's Wizard Wheezes through a secretive supplier, X. The demand for Wizard's Chess is given by:

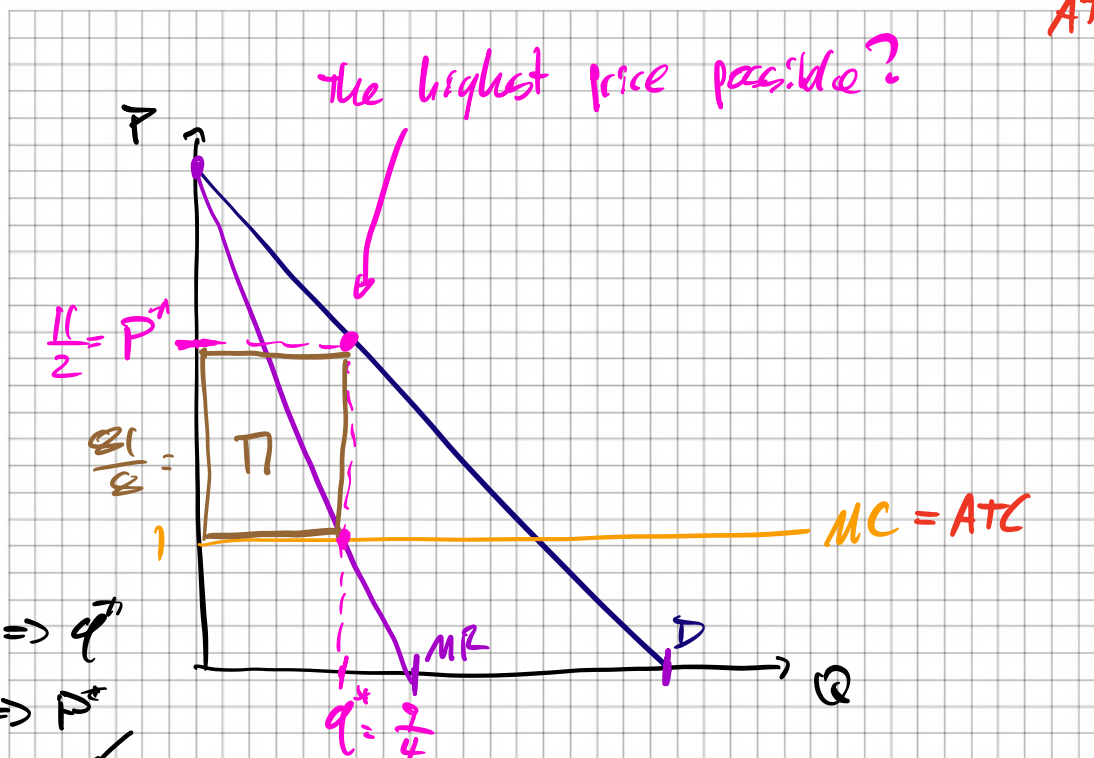
$$P = 10 - 2Q$$

The marginal cost of producing each game is a constant 1 galleon, and there are no fixed costs. The marginal revenue per game is:

$$1 = MC = MR = 10 - 4Q$$

$$TC = Q \times MC$$

$$ATC = MC$$



STEPS

1. $MC = MR \Rightarrow Q^*$
2. $Q^* \rightarrow D \Rightarrow P^*$
3. $\pi = Q^*(P - ATC)$

$$1 = 10 - 4Q \Rightarrow 4Q = 9 \Rightarrow Q = \frac{9}{4} = Q^*$$

Part A. Quantity

What is the profit maximizing number of games X should sell? Label on the graph above.

Part B. Price

What price should X charge per game? Label on the graph above.

$$P = 10 - 2 \cdot \left(\frac{9}{4}\right) = 10 - \frac{9}{2} = \frac{20 - 9}{2} = \frac{11}{2} = P^*$$

$$\pi = \frac{9}{4} \left(\frac{11}{2} - 1 \right) = \frac{9}{4} \cdot \frac{9}{2} = \frac{81}{8}$$

Part C. Profit

What is X's profit from selling the games? Label on the graph above.

Part D. Lumpsum Tax

$$MC = 10 - 4q = 0 \Rightarrow \bar{q} = \frac{10}{4} = \frac{5}{2} > \frac{9}{4}$$

As TikTok began to capture the attention of the magical world's youth, the Ministry of Magic implemented a 1 galleon subsidy on Wizard's Chess in an attempt to encourage more cognitively challenging pastimes. Use a graph to show the affect this subsidy had on the market.

